

Econometrics I

Lecture 11a: Matching

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Matching

Recall: Fundamental Problem of Causal Inference

Fundamental Problem of Causal Inference

- ▶ We don't observe the **counterfactual** $Y_i(1 - D_i)$.
- ▶ For a single individual we either observe $Y_i(1)$ **or** $Y_i(0)$ but never both!
- ▶ Matching says: Find someone with treatment status $1 - D_i$ but the same X_i (or **re-weight** your sample)
 - Assume all selection into treatment is based only on observed X_i
 - Ignore **selection on unobservables**

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$$

i	$Y_i(1)$	$Y_i(0)$	D_i	X_i
1	1	?	1	3.2
2	0	?	1	1.7
3	?	0	0	2.5
		$\vdots$		
n	?	1	0	-1.5

Suppose you ran a drug-trial experiment

- ▶ Collected demographic information on each patient (Age, Sex, Education, Income)
- ▶ Flipped a coin to determine treatment status $D_i = 1$ if *Heads*, $D_i = 0$ if *Tails*.
- ▶ First course of action: Construct a **covariate balance table**.
- ▶ If your control group is 70% male and your treatment group is 70% female, should you compare differences in means?
 - Probably we wanted to compare treated males to control males, and treated females to control females.
 - Otherwise, are we learning about the difference between sexes, or the effect of treatment?

Balance Table: Examples (see cobalt Package)

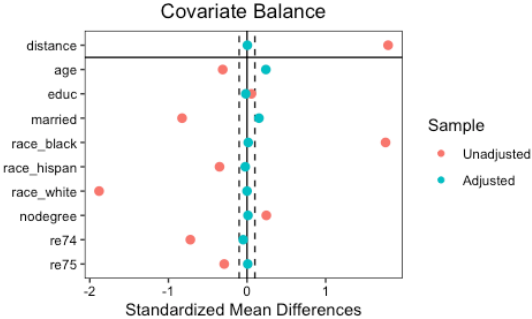


TABLE 1—BASELINE DATA

	Treatment (1)	Control (2)	Difference (3)
<i>Panel A. Teacher attendance</i>			
School open	0.66	0.64	0.02 (0.11)
	41	39	80
<i>Panel B. Student participation (random check)</i>			
Number of students present	17.71	15.92	1.78 (2.31)
	27	25	52
<i>Panel C. Teacher qualifications</i>			
Teacher test scores	34.99	33.54	1.44 (2.02)
	53	54	107
<i>Panel D. Teacher performance measures (random check)</i>			
Percentage of children sitting within classroom	0.83	0.84	0.00 (0.09)
	27	25	52
Percent of teachers interacting with students	0.78	0.72	0.06 (0.12)
	27	25	52
Blackboards utilized	0.85	0.89	-0.04 (0.11)
	20	19	39
F-stat (1,110)			1.21
p-value			(0.27)

Checking Balance: Standardized Differences

For each covariate k , compare the treated and control means in units of its standard deviation:

$$d_k = \frac{\bar{x}_{k,1} - \bar{x}_{k,0}}{\sqrt{(s_{k,1}^2 + s_{k,0}^2) / 2}}$$

- ▶ **Not a t -test.** A t -statistic grows with $\sqrt{n}$: trivial imbalance is “significant” in a big sample, and matching or trimming can raise p -values just by dropping observations.
- ▶ Rules of thumb: $|d_k| < 0.1$ is well balanced (Austin 2009); Imbens and Rubin (2015) worry above 0.25. Also check variance ratios $s_{k,1}^2 / s_{k,0}^2$: outside $[0.5, 2]$ is a warning.
- ▶ Report d_k before and after matching or weighting, for every covariate (and key squares and interactions). The left panel on the previous slide is exactly this: `cobalt::love.plot()`.
- ▶ Balance on observables is necessary, not sufficient: it says nothing about unobservables.

- Formally the key assumption is the **Conditional Independence Assumption (CIA)**

$$\{Y_i(1), Y_i(0)\} \perp D_i | X_i$$

- Once we know X_i allocation to treatment D_i is as if it is random.
- The only difference between treatment and control is composition $f(X_i | D_i)$ of the sample.
 - Re-weight so distribution of X_i is the same for both groups $X_i | D_i \sim X_i$ then we assume all other differences are irrelevant and can just compare means.
 - Unobservables that affect outcomes are **unrelated to treatment** conditional on X_i .

Simple Matching in Practice

For each observation i in the **treated group**, find $\mathcal{C}_i$ the set of observations in the **control group** with the same X_i :

$$ATT = \frac{1}{N_T} \sum_{i=1}^{N_T} \left(Y_i(1) - \frac{1}{|\mathcal{C}_i|} \sum_{j \in \mathcal{C}_i} Y_j(0) \right)$$

Some issues:

- ▶ For each observation $|\mathcal{C}_i|$ may be a different size (including zero!). This can make the variance a bit unpredictable.
- ▶ Finding **exact matches** gets harder as $\dim(X_i)$ grows.

Why would this work?

Let $F^1(x)$ be the distribution of characteristics in the treatment group, we can define the ATT as

$$\begin{aligned}\mathbb{E}[Y(1) - Y(0) \mid D = 1] &= \mathbb{E}_{F^1(x)} [\mathbb{E}(Y(1) - Y(0) \mid D = 1, X)] \text{ L.I.E.} \\ &= \mathbb{E}_{F^1(x)} [\mathbb{E}(\textcolor{brown}{Y}(1) \mid D = 1, X)] - \mathbb{E}_{F^1(x)} [\mathbb{E}(\textcolor{brown}{Y}(0) \mid D = 1, X)] \text{ linearity}\end{aligned}$$

The first part we observe directly:

$$= \mathbb{E}_{F^1(x)} [\mathbb{E}(Y(1) \mid D = 1, X)]$$

But the counterfactual mean is not observed!

$$= \mathbb{E}_{F^1(x)} [\mathbb{E}(Y(0) \mid D = 1, X)]$$

But conditional independence does this for us:

$$\mathbb{E}_{F^1(x)} [\mathbb{E}(Y(0) \mid \textcolor{brown}{D} = 1, X)] = \mathbb{E}_{F^1(x)} [\mathbb{E}(Y(0) \mid \textcolor{brown}{D} = 0, X)]$$

(For the ATT we only need $Y(0) \perp D \mid X$ and $P(X) < 1$.)

- ▶ Think about

$$Y_i = \alpha + \beta D_i + u_i$$

- ▶ Assume that $\mathbb{E}(u_i \mid D_i, X_i) = \mathbb{E}(u_i \mid X_i) = \gamma_0 + \gamma X_i$ (conditional mean independence with a linear control function)
- ▶ Then we can get β consistently (but not other variables) by running the following:

$$Y_i = \alpha + \beta D_i + \gamma X_i + v_i$$

- ▶ Again we are in the homogeneous treatment world

So matching works in dimension 1. But what if $\dim(X) > 1$?

- ▶ True high-dimensional matching may be infeasible: there may be no controls with the same X_i , so we cannot build weights $w(x)$ with $f(x \mid D = 1) = w(x) f(x \mid D = 0)$ from the sample.
- ▶ One solution is the nearest-neighbor approach in Abadie and Imbens (2006) (later on).
- ▶ Suppose instead we had a **sufficient statistic**

- ▶ Rosenbaum and Rubin (1983) propose the **propensity score** as a sufficient statistic

$$\mathbb{P}(D_i = 1 \mid X_i) \equiv P(X_i)$$

- ▶ Idea: avoid high-dimensional problems by matching on $P(X_i)$.
- ▶ They prove that the propensity score and any function of X , $b(X)$ which is finer serves as a **balancing score**.
- ▶ Finer implies that:

$$\begin{aligned} b(X^1) = b(X^2) &\implies P(X^1) = P(X^2) \\ P(X^1) = P(X^2) &\not\Rightarrow b(X^1) = b(X^2) \end{aligned}$$

- Main Result: If treatment assignment is strongly ignorable conditional on X (CIA) then it is strongly ignorable $Y(1), Y(0) \perp D \mid b(X)$ given any balancing score $b(X)$ including the propensity score:

$$\begin{aligned}\mathbb{P}(D = 1 \mid Y(1), Y(0), P(X)) &= \mathbb{E}[\mathbb{P}(D = 1 \mid Y(1), Y(0), X) \mid Y(1), Y(0), P(X)] \\ &= \mathbb{E}[P(X) \mid Y(1), Y(0), P(X)] = P(X) \quad (\text{CIA in the second})\end{aligned}$$

- Also we require that $0 < P(X) < 1$ at each X which is known as the **support condition**.
- The theorem implies that given $P(X)$ we have as if random assignment.

Just like in the matching case the problem arises because we do not observe the counterfactual mean:

$$\mathbb{E}_{F^1(x)}[\mathbb{E}(Y(0) \mid D = 1, X)]$$

With conditional independence and the propensity score:

$$\begin{aligned}\mathbb{E}_{F^1(x)}[\mathbb{E}(Y(0) \mid D = 1, X)] &= \mathbb{E}_{F^1(x)}[\mathbb{E}(Y(0) \mid D = 0, X)] \\ &= \mathbb{E}_{F^1(x)}[\mathbb{E}(Y(0) \mid D = 0, P(X))]\end{aligned}$$

- ▶ Instead of matching on high dimensional X we can now match on a one-dimensional propensity score.
- ▶ Thus the propensity score provides **dimension reduction**
- ▶ We still have to estimate the propensity score which is a high dimensional problem!
 - An easy way would be to use $\pi(x) = P(x)$ from logit or probit.
 - To be fancy, include polynomial interactions in X_i .
- ▶ People often match on **caliper** of propensity score $d_{ij} = |\pi(x_i) - \pi(x_j)|$ where $d_{ij} < c$
 - Compare to other x_j with similar $\pi(x_i) = 0.34$ and $\pi(x_j) = 0.32$.

How do we implement?

- Kernels are an obvious choice

$$\widehat{ATT} = \frac{1}{N_1} \sum_{i:D_i=1} \left[Y_i - \frac{\sum_{j:D_j=0} Y_j \cdot \mathcal{K} \left(\frac{P(X_i) - P(X_j)}{h} \right)}{\sum_{s:D_s=0} \mathcal{K} \left(\frac{P(X_i) - P(X_s)}{h} \right)} \right]$$

where N_1 is the sample size of the treatment group

and $\mathcal{K}(u)$ is a valid Kernel weight with bandwidth h (people tend to use Gaussian Kernels here)

- As your propensity score gets further away from observation i you get less weight
- As $h \rightarrow 0$ (or σ_h) the window gets smaller and we use fewer neighbors.

- Calculate a smoothed estimate of the treatment probability $\pi(x) = \mathbb{P}(D_i = 1 \mid x)$. (Logit or Probit)

$$ATE = \frac{1}{n} \sum_{t \in \text{Treatment}} \frac{y_t}{\pi(x_t)} - \frac{1}{n} \sum_{s \in \text{Control}} \frac{y_s}{1 - \pi(x_s)}$$

- How to get $\pi(x)$? Run a logit or probit.
- Normalize the weights within each arm (Hájek):

$$\widehat{ATE} = \frac{\sum_i D_i Y_i / \pi(x_i)}{\sum_i D_i / \pi(x_i)} - \frac{\sum_i (1 - D_i) Y_i / (1 - \pi(x_i))}{\sum_i (1 - D_i) / (1 - \pi(x_i))}$$

- Helps with very large weights; also consider trimming to $0.1 \leq \pi(x) \leq 0.9$ (Crump et al. 2009).

Doubly Robust: Augmented IPW (AIPW)

Combine an outcome model $\hat{\mu}_d(x) = \hat{\mathbb{E}}[Y \mid D = d, X = x]$ with the propensity score:

$$\widehat{ATE}_{AIPW} = \frac{1}{n} \sum_{i=1}^n \left[\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{D_i (Y_i - \hat{\mu}_1(X_i))}{\hat{\pi}(X_i)} - \frac{(1 - D_i) (Y_i - \hat{\mu}_0(X_i))}{1 - \hat{\pi}(X_i)} \right]$$

- ▶ A regression estimate plus an IPW correction built from its residuals.
- ▶ **Doubly robust:** consistent if *either* $\hat{\pi}$ or $\hat{\mu}_d$ is correctly specified. If $\hat{\mu}_d$ is right, the correction has mean zero; if $\hat{\pi}$ is right, the correction removes $\hat{\mu}_d$'s bias.
- ▶ With both right, it attains the semiparametric efficiency bound (Robins, Rotnitzky and Zhao 1994; Hahn 1998).
- ▶ Still needs CIA and overlap: it does not fix unobserved confounding, and small $\hat{\pi}$ still means big weights.
- ▶ The building block of double/debiased ML (Chernozhukov et al. 2018): flexible π, μ_d with cross-fitting. In R: `grf::average_treatment_effect()`, `DoubleML`.

IPW vs Propensity Score: What's the Difference?

Propensity Scores

- ▶ Find individual with similar $p(X)$
- ▶ Excludes unmatched participants
- ▶ Creates a balanced **sub-sample**
- ▶ Can be sensitive to the choice of matching algorithm
- ▶ With enough data can compute $\mathbb{E}[Y_i(1) - Y_i(0) \mid P(X_i)]$

Inverse Probability Weights

- ▶ Reweights the entire sample
- ▶ Keeps all participants
- ▶ Creates a pseudo-population
- ▶ Can be sensitive to extreme weights
- ▶ We only compute ATE or ATT

Other approaches in high dimensions

- ▶ Given two vectors $\mathbf{x}$ and $\mathbf{y}$, how to choose $d(\mathbf{x}, \mathbf{y})$ the distance function?
- ▶ Papers mostly use Mahalanobis distance: $d(\mathbf{x}, \mathbf{y}) = (\mathbf{x} - \mathbf{y})\Sigma^{-1}(\mathbf{x} - \mathbf{y})'$.
 - Quadratic distance with inverse covariance matrix as “weights”.
 - Reduces to Euclidean distance when $\Sigma = I$ (standardized Euclidean for diagonal Σ).
- ▶ Older papers use caliper matching anything within $\|\mathbf{x}_s - \mathbf{x}_t\| < b$ is a match
 - Number of matches varies from observation to observation.
 - Variance can be unpredictable: some $(\mathbf{y}_i, \mathbf{x}_i)$ have many matches, others have none
 - Some obs may have nothing within $\|\mathbf{x}_s - \mathbf{x}_t\| < b$? Drop these?
 - Probably avoid this unless you have a good reason...
- ▶ Let's explore k Nearest Neighbor Matching (kNN)

Nonparametric k -NN Matching: Abadie and Imbens (2006)

For each observation i , we observe $Y_i(D_i)$; compute a **counterfactual** $\hat{Y}_i(1 - D_i)$:

$$\begin{aligned}\hat{Y}_i(0) &= \begin{cases} Y_i & \text{if } D_i = 0 \\ \frac{1}{\#\mathcal{J}_M(i)} \sum_{l \in \mathcal{J}_M(i)} Y_l & \text{if } D_i = 1 \end{cases} \\ \hat{Y}_i(1) &= \begin{cases} \frac{1}{\#\mathcal{J}_M(i)} \sum_{l \in \mathcal{J}_M(i)} Y_l & \text{if } D_i = 0 \\ Y_i & \text{if } D_i = 1 \end{cases}\end{aligned}$$

- ▶ $\#\mathcal{J}_M(i)$ is the number of matches for i of opposite treatment assignment $D_l = 1 - D_i$.
- ▶ $d_M(i)$ is the distance to the M -th closest unit with $D_l = 1 - D_i$;
 $\mathcal{J}_M(i) = \{l : D_l = 1 - D_i, \|X_l - X_i\| \leq d_M(i)\}$.
- ▶ If there are ties $\#\mathcal{J}_M(i) > M$.
- ▶ This is just k -NN matching.

Each observation i gets a weight based on how often it is used as a match for other observations l :

$$K_M(i) = \sum_{l=1}^N 1\{i \in \mathcal{J}_M(l)\} \frac{1}{\#\mathcal{J}_M(l)}$$

Observations used in lots of matches get more weight. $\sum_{i=1}^N K_M(i) = N$.

This is just a weighted average of Y_i values (aka a **kernel**!):

$$ATE_M = \frac{1}{N} \sum_{i=1}^N [\widehat{Y}_i(1) - \widehat{Y}_i(0)] = \frac{1}{N} \sum_{i=1}^N \underbrace{(2D_i - 1) [1 + K_M(i)]}_{w_{i,M}} Y_i$$

Different weighting schemes give different parameters:

$$ATE_M = \frac{1}{N} \sum_{i=1}^N (2D_i - 1) [1 + K_M(i)] \cdot Y_i$$

$$ATT_M = \frac{1}{N_1} \sum_{i=1}^N [D_i - (1 - D_i) \cdot K_M(i)] \cdot Y_i$$

$$ATUT_M = \frac{1}{N_0} \sum_{i=1}^N [D_i \cdot K_M(i) - (1 - D_i)] \cdot Y_i$$

Nonparametric k -NN Matching: Bias Correction (Abadie and Imbens 2011)

We can use weighted least squares to adjust the predictions $\hat{Y}_i(D_i)$:

$$(\hat{\beta}_{t,0}, \hat{\beta}_{t,1}) = \operatorname{argmin}_{\{\beta_{t,0}, \beta_{t,1}\}} \sum_{i:D_i=t} K_M(i) (Y_i - \beta_{t,0} - \beta'_{t,1} X_i)^2$$

Where $\hat{\mu}_t(x) = \hat{\beta}_{t,0} + \hat{\beta}'_{t,1}x$ are the fitted regression functions for treatment ($t = 1$) and control ($t = 0$).

$$\tilde{Y}_i(0) = \begin{cases} Y_i & \text{if } D_i = 0 \\ \frac{1}{\#\mathcal{J}_M(i)} \sum_{l \in \mathcal{J}_M(i)} \{Y_l + \hat{\mu}_0(X_i) - \hat{\mu}_0(X_l)\} & \text{if } D_i = 1 \end{cases}$$
$$\tilde{Y}_i(1) = \begin{cases} \frac{1}{\#\mathcal{J}_M(i)} \sum_{l \in \mathcal{J}_M(i)} \{Y_l + \hat{\mu}_1(X_i) - \hat{\mu}_1(X_l)\} & \text{if } D_i = 0 \\ Y_i & \text{if } D_i = 1 \end{cases}$$

So that the ATE is given by: $ATE_M = \frac{1}{N} \sum_{i=1}^N \{\tilde{Y}_i(1) - \tilde{Y}_i(0)\}$

- ▶ Base case is easy enough to do on your own using a canned k -NN package.
- ▶ Even regression/bias adjustment is pretty easy
- ▶ Computing ATE, ATT, ATUT is relatively straightforward on the matched sample
- ▶ MatchIt in R (Ho, Imai, King and Stuart) or nnmatch in Stata (Abadie et al.)
- ▶ Documentation here is pretty good:

https://scholar.harvard.edu/files/imbens/files/sjpdf-1.html_.pdf.

Case Study

A Matching Example

Here is an example where I found that matching was helpful in my own work with Julie Mortimer:

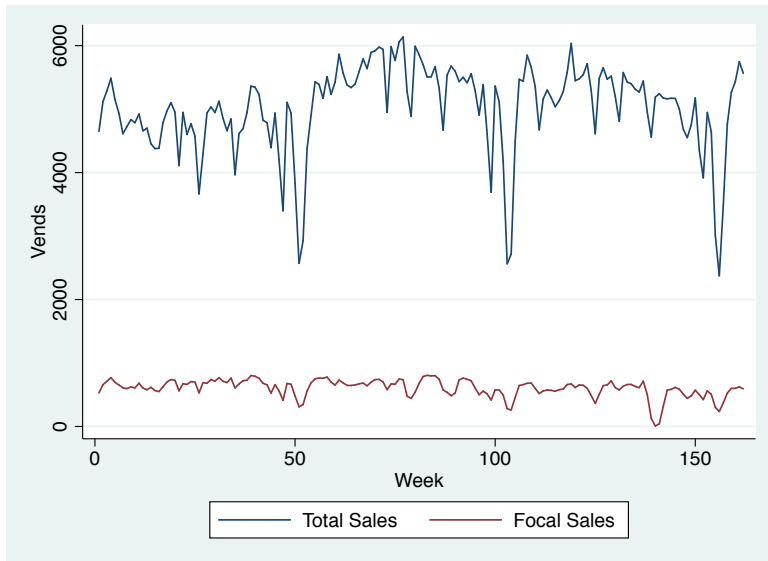
- ▶ We ran a randomized experiment where we removed Snickers bars from around 60 vending machines in office buildings in downtown Chicago.
- ▶ We have a few possible control groups:
 1. Same vending machine in other weeks (captures heterogeneous tastes in the cross section)
 2. Other vending machines in the same week (might capture aggregate shocks, ad campaigns, etc.)
- ▶ We went with #1 as #2 was not particularly helpful.

A Matching Example

Major problem was that there was a ton of heterogeneity in the overall level of (potential) weekly sales which we call M_t .

- ▶ Main source of heterogeneity is how many people are in the office that week, or how late they work.
- ▶ Based on total sales our average over treatment weeks was in the 74th percentile of all weeks.
- ▶ This was after removing a product, so we know sales should have gone down!
- ▶ How do we fix this without running the experiment for an entire year?
- ▶ Can't use shares instead of quantities. Why?

Sales over Time



A Matching Example

Ideally we could just observe M_t directly and use that as our matching variable X

- ▶ We didn't observe it directly and tried a few different measures:
 - Sales at the soda machine next to the snack machine
 - Sales of salty snacks at the same machine (not substitutes for candy bars).
 - We used k-NN with $k = 4$ to select control weeks – notice we re-weight so that overall sales are approximately the same (minus the removed product).
- ▶ We also tried a more structured approach:
 - Define control weeks as valid IFF
 - Treatment-week total sales are weakly lower than the control week's total
 - but no lower than the control week's total minus its Snickers sales.

Product	Control Mean	Control %ile	Treatment Mean	Treatment %ile	Mean Difference	% Δ
<i>Vends</i>						
Peanut M&Ms	359.9	73.6	478.3*	99.4	118.4*	32.9
Twix Caramel	187.6	55.3	297.1*	100.0	109.5*	58.4
Assorted Chocolate	334.8	66.7	398.0*	95.0	63.2*	18.9
Assorted Energy	571.9	63.5	616.2	76.7	44.3	7.8
Zoo Animal Cracker	209.1	78.6	243.7*	98.1	34.6*	16.5
Salted Peanuts	187.9	70.4	216.3*	93.7	28.4	15.1
Choc Chip Famous Amos	171.6	71.7	193.1*	95.0	21.5*	12.5
Ruger Vanilla Wafer	107.3	59.7	127.9	78.6	20.6*	19.1
Assorted Candy	215.8	43.4	229.6	60.4	13.7	6.4
Assorted Potato Chips	279.6	64.2	292.4*	66.7	12.8	4.6
Assorted Pretzels	548.3	87.4	557.7*	88.7	9.4	1.7
Raisinets	133.3	66.0	139.4	74.2	6.1	4.6
Cheetos	262.2	60.1	260.5	58.2	-1.8	-0.7
Grandmas Choc Chip	77.9	51.3	72.5	37.8	-5.4	-7.0
Doritos	215.4	54.1	203.1	39.6	-12.3*	-5.7
Assorted Cookie	180.3	61.0	162.4	48.4	-17.9	-10.0
Skittles	100.1	62.9	75.1*	30.2	-25.1*	-25.0
Assorted Salty Snack	1382.8	56.0	1276.2*	23.3	-106.7*	-7.7
Snickers	323.4	50.3	2.0*	1.3	-321.4*	-99.4
Total	5849.6	74.2	5841.3	73.0	-8.3	-0.1

Notes: Control weeks are selected through nearest-neighbor matching using four control observations for each treatment week. Percentiles are relative to the full distribution of control weeks.

- ▶ This would be a good time to work through the vignette for cobalt
<https://cran.r-project.org/web/packages/cobalt/vignettes/cobalt.html>
- ▶ Compare the ATT for the LaLonde data with the IPW, Nearest Neighbor, and Propensity Score estimates.
- ▶ Then start the homework